

8 A library of analysis tools

Data analysis is a mandatory step in our work and to have an efficient, well tested library of tools is very important. Between 2014 and 2016, a quite basic but complete set of standard tools was developed in C++, in order to achieve good performance. Basic analysis of typical Monte Carlo data has been implemented using [17, 18] books as reference. The main features available are the calculation of mean, variance, skewness and kurtosis of a given set of *correlated* data, with a correct estimate of the correspondent statistical error using either the Jackknife or the bootstrap method (cf. also section 3.3 of [14]). Another important analysis algorithm which has been implemented in a very general fashion is the multiple histogram method, also known as Ferrenberg-Swendsen reweighting (cf. section 8.2 of [17] and sub-section 3.4.4 of [14]). A couple of executables have been, then, written for the user, in order to comfortably use the tools offered. Clearly, you could use the code as external library for another project, but this should not be necessary.



`/home/phil-shared/Codebases/DataAnalysisTools.git`

As usual, this code base can be obtained cloning a Git repository. Unfortunately, at the moment, no documentation is available and you need to explore the repository to understand what has been done how. However, as far as building the code is concerned, the procedure is quite standard and similar to that of CL²QCD via CMAKE. After having cloned the repository, create a `build` folder in it and move into it. Here, you can simply run `cmake .` in order to run a default setup or you can use the `ccmake` command to have access to configuration options. If no error occurs (in case have a look to the error and act accordingly), you can run `make`, optionally using the `-j` option to use all the cores of the PC. At this stage there should be no compilation error and a couple of executable should be created (also tests executable are built, so that you can run the code base tests using `ctest`). Use

`dataAnalysis` to calculate moments of observables given some data files;

`lqcdReweighting` to reweight Monte Carlo observables and/or their moments.

Every executable accepts command line options and you can get some documentation running the executables with the `-h` option.